

Concept Review

Thermocouples

Why Thermocouples?

Temperature measurement is fundamental to virtually every engineering discipline, and thermocouples are the most widely used industrial temperature sensors in the world due to their versatility, ruggedness, and cost-effectiveness. They are found everywhere, from research laboratories to manufacturing processes and control systems. Real-world applications range from monitoring jet engine exhaust exceeding 1000°C to controlling plastic injection molding with precision accuracy.

Understanding their operating principles, junction styles, and selection criteria enables engineers to specify the right sensor for each application, balancing response time, accuracy, durability, and cost to achieve optimal performance in temperature monitoring and control systems.

1. Introduction



A thermocouple is an electrical device consisting of two dissimilar metal conductors joined at one end to form an electrical junction. When this junction experiences a temperature difference from the other end of the wires, it produces a small voltage (in millivolts) that can be measured and interpreted as temperature. This phenomenon is known as the **Seebeck effect**, which describes how thermal energy or temperature is converted into electrical energy when two different conductors are connected.

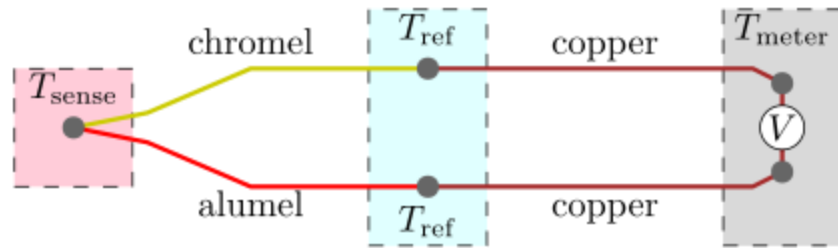
Thermocouples are among the most widely used temperature sensors in industry and research. Along their simplicity, durability, and wide operating ranges, they offer several unique advantages: they are self-powered (requiring no external excitation), can measure extremely high and low temperatures, and are relatively inexpensive. However, their main limitation is accuracy, where achieving system errors of less than one degree Celsius can be challenging.

2. How Thermocouples Work

The operating principle of a thermocouple is straightforward:

1. Two specially selected dissimilar metal wires are joined at one end, creating the **hot junction** (also called the measuring point)
2. The other ends of the wires remain separated and are held at a **cold junction** (also called the reference junction or connection point)
3. When the hot junction is exposed to the temperature being measured, heat conducts into the conjoined metals
4. The temperature difference between the hot and cold junctions causes electrons to flow differently in each metal due to their unique electrical properties
5. This varying electron density creates a small voltage that can be measured with a voltmeter

The key insight is that thermocouples measure the **differential temperature** between the hot junction and the cold junction, not absolute temperature. The measured voltage varies linearly with temperature, and the two metals are specifically paired for their ability to generate a millivoltage output that correlates to temperature changes. This measured voltage is used to calculate the temperature based on the reference voltage at the cold junction.



a. Cold Junction Compensation

Since thermocouples measure temperature difference rather than absolute temperature, accurate readings require knowing the temperature at the cold junction. This is where **cold junction compensation (CJC)** becomes essential.

In modern thermocouple systems, a semiconductor temperature sensor (such as the MAX31855 interface chip located in the Mechatronic Sensors Trainer) measures the ambient temperature at the cold junction. The system then measures the voltage from the thermocouple output at the reference junction, converts this to an uncompensated temperature value and adds the cold junction temperature to calculate the actual hot junction temperature.

This compensation happens in dedicated thermocouple interface chips, making implementation with microcontrollers quick and straightforward. The MAX31855, for example, features a 14-bit ADC, offering 0.25°C resolution with $\pm 2^\circ\text{C}$ accuracy for K-type thermocouples, reading temperatures between -200°C and $+700^\circ\text{C}$.

3. Types of Thermocouples

Different pairs of metals produce different voltage-temperature characteristics. Each thermocouple type has been assigned a letter designation that refers to its specific metal pairing and calibration.

Type K is the industry standard due to its wide temperature range, good accuracy, reliability, and relatively low cost. **Type J** thermocouples use iron as one conductor, making them unsuitable for moist environments due to corrosion risk. **Type T** thermocouples excel at measuring extremely low temperatures in cryogenic applications. Here are the most common types:

Type	Metal Pairing	Temperature Range	Common Applications
K	Nickel-Chromium / Nickel-Alumel	-200°C to $+1,260^\circ\text{C}$	General purpose, most popular
J	Iron / Constantan	-40°C to $+750^\circ\text{C}$	Plastics, vacuum applications (not for moist environments)
T	Copper / Constantan	-200°C to $+350^\circ\text{C}$	Low temperature, cryogenic processes

N	Nicrosil / Nisil	-270°C to +1,300°C	High-temperature stability, oxidation resistance
E	Nickel-Chromium / Constantan	-270°C to +1,000°C	High output, good accuracy
R, S	Platinum-Rhodium alloys	0°C to +1,600°C	High-precision, expensive

4. Thermocouple Probes and Junction Styles

When the thermocouple wires and their junction are inserted into a protective metal tube called a **sheath** (commonly stainless steel), it is called a **thermocouple probe**. The wires inside the sheath do not touch except at the measuring junction.

There are 3 types of ways in which the junction and the sheath can come together, each with distinct performance characteristics:

- **Grounded Junction:** The junction tip is physically welded to the internal end of the sheath, creating metal-to-metal contact. This design offers: faster response time than ungrounded junctions due to efficient heat transfer through the metal pathway; protection from corrosive environments and high pressures; and electrical connection to the sheath, which can create susceptibility to ground loop interference.

Grounded junction



Ungrounded junction



Exposed junction



- **Ungrounded Junction:** The junction is enclosed within the sheath but electrically isolated from it, typically surrounded by a nonconductive material. This configuration has the best accuracy and due to the electrical isolation, it is ideal for interfacing with microprocessors and control systems. However, it has the slowest response time due to the insulation material between the junction and sheath.

- **Exposed Junction:** The measuring junction is completely unprotected and in direct contact with the process being measured. This provides the fastest response time since heat transfers directly to the junction. However, exposed junctions are fragile and susceptible to corrosion, making them unsuitable for harsh industrial environments.

You can verify if your thermocouple is grounded or ungrounded by using a voltmeter to check for conductivity between the sheath and one of the 2 leads on the connector end. If there is conductivity, it has a grounded junction.

a. Response Time Considerations

A sensor's **time constant** is defined as the time required to reach 63.2% of a step change in temperature under specified conditions. Response time is affected by the probe diameter (larger diameter = slower response), the wire gauge (thicker wire = slower response) and the junction style (exposed > grounded > ungrounded)

5. Applications and Uses

Thermocouples are utilized across virtually every industry requiring temperature measurement, some examples are:

Industrial Processes: Temperature measurement in kilns, gas turbine exhaust, diesel engines, metal heat treatment, ceramics, and glass manufacturing. Types K and N are especially common in high-temperature applications due to their stability and oxidation resistance.

HVAC Systems: Monitoring supply and return air temperatures, boiler and chiller performance, and duct temperature profiling in commercial and industrial facilities.

Food Processing: Ensuring product safety during cooking, pasteurization, freezing, and storage due to their fast response times and sanitary probe configurations.

Plastics Manufacturing: Monitoring barrel, nozzle, and mold temperatures in injection molding processes to maintain material quality and prevent defects.

Residential Applications: Temperature sensors in thermostats and flame sensors in safety devices for gas-powered appliances.

6. Datasheet And Considerations

Selecting the appropriate thermocouple involves considering five key factors:

1. Application Requirements: Determine where and how the sensor will be used. Will it measure gas, liquid, or solid temperatures? Is it for laboratory or industrial use?

2. Temperature Range: Consult thermocouple range charts to match your expected temperature range with the appropriate type. Type K works for most general applications, Type N for extreme high temperatures, and Type T for cryogenic conditions.

3. Response Time: If fast response is critical, choose an exposed junction. For corrosive environments requiring reasonable speed, select a grounded junction. When accuracy and electrical isolation are paramount, choose an ungrounded junction.

4. Environmental Conditions: Consider chemical resistance, abrasion, vibration, and pressure. Exposed junctions are limited to calm environments, while grounded and ungrounded junctions can handle harsh conditions.

5. Installation Requirements: Ensure compatibility with existing equipment, including probe diameter, connector types, and mounting constraints.

a. Comparison with Other Temperature Sensors

i. Advantages:

- Self-powered (no external excitation required)
- Simple, rugged construction
- Wide temperature ranges (absolute zero to beyond melting point of steel)
- Inexpensive and readily available
- Fast response times possible
- Not affected by lead wire resistance

ii. Disadvantages:

- Lower accuracy than RTDs and thermistors
- Less stable than RTDs over time
- Requires cold junction compensation
- Low voltage output susceptible to electrical noise
- Output requires linearization for precise readings

Comparison Summary: Thermocouples generally measure temperatures over wider ranges and are more rugged than RTDs (Resistance Temperature Detectors) and thermistors, but are less accurate. RTDs offer better stability and accuracy but are more expensive and subject to self-heating errors. Thermistors provide the best accuracy but have limited temperature ranges.

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